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Contents

Abbreviations

Acronym / Abbreviation	Meaning
DR	Drug repurposing
DTI	Drug-target interactions
MSD	Multiple sulfatase deficiency
AI	Artificial intelligence
ML	Machine learning

Abstract

Work package 4 (WP4) of REMEDi4ALL focuses on surveying and developing open-source artificial intelligence (AI) tools and databases for supporting user-driven *in silico* drug repurposing applications. Starting with a landscape analysis, Task 4.1 surveyed 196 *in silico* resources (tools, methods, and databases) in terms of their usefulness and reliability for various drug repurposing applications. For each resource, we provide a comprehensive list of annotations, such as open-data statistics and open-implementation of the databases and tools, respectively, and a summary ranking of the resources among 26 experts of the drug repurposing field. The survey results have been made available as a sustainable and extendable catalogue online at https://idrc-r4a.com/. Furthermore, Task 4.1 carried out three demonstrator use cases using selected *in silico* resources to guide the drug repurposing process in the demonstrators of the REMEDi4ALL project. This comprehensive survey and expert evaluation demonstrate the benefits of using *in silico* resources for drug repurposing and highlights some limitations and missing gaps in the current resources.

1. Introduction

The development of a new drug requires a significant amount of time and investment costs (Dickson & Gagnon, 2009). Since the number of new drug approvals has become stagnated (Dowden & Munro, 2019) (Batta et al., 2020), due to this and other factors, finding new uses for approved drugs or investigational/shelved compounds (so-called drug repurposing or repositioning) has become a popular alternative strategy for drug development (Shaughnessy, 2011), mainly because of it is cost- and time-efficiency (Pushpakom et al., 2019)(Gupta et al., 2013a).

In oncology, DR can be further divided into anticancer drugs that are used off-label and drugs that have been developed as anticancer or treatments for other indications but were discarded or deprioritized due to lack of efficacy or dose-limiting toxicity (Schipper et al., 2022). In the latter case, when drugs are used completely outside their intended indications, the benefits of DR may be reduced if the cancer treatment requires different dosing, such as methotrexateas an anticancer drug. Therefore, there are still many challenges on the road to clinical use of DR candidates, related to, for instance, safety and pharmacokinetic properties, especially when moving from monotherapies to drug combinations.

Over recent years, various computational resources have been developed to support systematic DR efforts. Popular information sources for *in-silico* DR include, for instance, electronic health records, genome-wide association analyses and gene expression response profiles, pathway mappings, compound structures, target binding assays and other types of phenotypic profiling data, as well as toxicity and other clinical study data (Ji et al., 2019; Lin et al., 2019; Pushpakom et al., 2019; Verbaanderd et al., 2017; Wu et al., 2019). Several original studies and review articles have introduced new approaches and compared various resources directly or indirectly related to drug repositioning (Corsello et al., 2017) (Shameer et al., 2018). For instance, in the review by Dudley et al. on computational methods for drug repositioning, the authors classified the methodologies either as drug or disease-based repositioning (Dudley et al., 2011).

Jin et al. linked the existing drug repositioning methods with their integrated biological and pharmaceutical knowledge and discussed how to customize new drug repositioning pipelines for specific indications (Jin & Wong, 2014). Sam et al. surveyed several web-based tools that can aid DR (Sam & Athri, 2019), while Song et al. discussed various tools and resources that are developed for DR (Song et al., 2009). Li et al. classified recent progress in computational drug repositioning into four parts: repositioning strategies, computational approaches, validation methods, and application areas (Li et al., 2016). Recently, Yang et al. extensively reviewed using artificial intelligence (AI) for DR (Yang et al., 2019). Similarly, Tanoli et al. reviewed 102 databases by dividing those into four main categories and 27 subcategories (Tanoli et al., 2020), as well as web-based tools and AI-based methods for DR (Tanoli et al., 2021).

Even though hundreds of tools, databases and methods are already available for drug repurposing, several new resources are being developed each year to support drug repurposing efforts. The primary challenge in drug repurposing is that the mechanistic and phenotypic information needed to distil new targets or indications for the existing drugs is spread out over a vast and increasing number of databases and web platforms, and these data and predictions may vary significantly in coverage, quality, and reliability.

To provide an updated information for the REMEDi4ALL project, we surveyed 196 in silico resources for supporting drug repurposing, including web-tools, open-access databases and prediction methods, and divided those into eleven categories based on the data source (e.g., cell lines or patient samples), and DR applications (e.g., target profiles and biological pathways, drug structure, toxicity and disease applications). Detailed annotations for these surveyed 196 resources have been made publicly available for the community online at: https://idrc-r4a.com/. Furthermore, we ranked the *in-silico* resources in each category based on scoring provided by 26 experts in the field, summarizing the resources based on several factors, including comprehensiveness, data quality, resource availability, reliability and reusability, and the updating frequency of the resource. Finally, we demonstrate the usage of the resources in selected uses cases of the REMEDi4ALL project to demonstrate the practical application of the *in-silico* resources in DR process.

2. Methods

2.1. Classification of 196 resources

To make a survey and catalogue of dozens of in silico resources available, we first defined a unique DR ontology that hierarchically divides the resources into categories, subcategories and sub-subcategories at three levels. At level 1 we divided the resources into 'phenotypic' and 'target-based' categories; level 2 contains sub-categories such as: 'Genomics', 'Docking',' Biochemical', 'Cell based', 'Drug-disease links, 'Toxicities', and 'Patient data'. Level 3 finally divides the resources into 15 sub-subcategories based on DR applications and the data types involved in the particular resource. For instance, Genomics subcategory was divided into two sub-subcategories i.e., 'Cancer mutations' and 'Gene-disease links'. Similarly, docking was divided into five sub-subcategories i.e., 'Protein structures', 'Docking', 'Homology modelling', 'Binding sites', and 'Chemical structures'. Biochemical was divided into two sub-subcategories i.e., 'Drug-target interactions' and 'Pathways'. "Cell-based" was further divided into three sub-subcategories i.e., 'Cell line data, 'Combination web tools', and 'Combination databases'. Besides these 15 sub-subcategories, we made three additional broader-level sub-subcategories (i.e., 'Web-based prediction tools', 'Prediction methods' and 'GUI-based data-mining tools'). These general sub-subcategories are not specific to a particular DR application but can be used in many applications.

After finalising these pre-defined categories, subcategories and sub-subcategories, we populated and surveyed selected resources belonging to each sub-subcategory, with the help of 26 experts in the DR field (please see the list under *Other contributors* on page 1 and ANNEX I). The first survey resulted in a total of 196 in silico resources across the (sub)categories, which were known and used by the experts. Furthermore, we defined a unique set of annotations/columns for each sub-subcategory, as shown in the online catalogue at https://idrc-r4a.com/. Such a comprehensive DR ontology and online catalogue will help to continuously update the catalogue, when new DR resources become available under various categories, and what annotations are required for resources in that sub-subcategory, respectively. The DR ontology can also help researchers who survey resources in other fields to define ontologies of similar type.

2.2. Expert ranking of the 196 resources

After the categorization of the resources, 26 experts (both REMEDi4ALL partners and external experts) ranked these resources by scoring between 0-10 (0: worst, 10: best) in terms of their usability, quality of data, and range of supported diseases. Ranking was based on several factors, such as data statistics, resource last update and availability, data availability (public/propriety), whether the resource has an API, disease specific or agnostic. In case of multiple rankings for the same resource, we took the median score to evaluate the resource. During the survey, we also annotated the resources using various classes, including clinical/preclinical, disease-specific/agnostic, pre/post-screening, open source and data/proprietary, supporting FAIR principles.

3. Results

3.1. Resources in each sub-subcategory

The sub-subcategory: ‘Drug-target interactions’ contains 26 databases, and ChEMBL was ranked as the top database. The combinations databases sub-subcategory contains two databases, and DrugComb was ranked as the top database. Combination web tools contain two web prediction tools, and SynergyFinder was ranked as the top database in this category. Nine tools were surveyed under the sub-subcategory ‘Homology modelling’, and Alphafold was ranked as the top tool. Twenty-three tools were surveyed under the sub-subcategory ‘Docking’, and Autodoc Vina was ranked at the top. The binding sites sub-subcategory comprises seven tools, and Chimera was ranked at the top. The sub-subcategory ‘Protein structures’ contains four databases and PDB was the top-ranked database. Three databases were surveyed in the sub-subcategory ‘Chemical structures’, and ChemSpider was at the top. The top database in the sub-subcategory ‘Patient responses’ was cBioPortal among 8 surveyed databases. The cell line response sub-subcategory contains 21 databases, and DepMap was at the top. The top database in the sub-subcategory: ‘Drug-disease links’ was ClinicalTrials.gov among the 12 surveyed databases. Three databases were surveyed in ‘Cancer mutations’, and COSMIC was at the top. OpenTargets was ranked as the top database in ‘Gene-disease links’ among the five surveyed databases. Reactome was ranked the top database in the sub-subcategory: ‘Pathways’ among ten surveyed databases. Finally, Comparative Toxicogenomics Database (CTD) was ranked the top database in the sub-subcategory: ‘Toxicities’ from the nine surveyed databases. 95% of the resources were either open-access (no sensitive data) or open-source (codes for reusability). Figure 1 shows the hierarchical classification, number of resources and the top-resource in each of the 15 sub-subcategories, whereas more detailed annotations of the individual resources have been made available online at: https://idrc-r4a.com/.

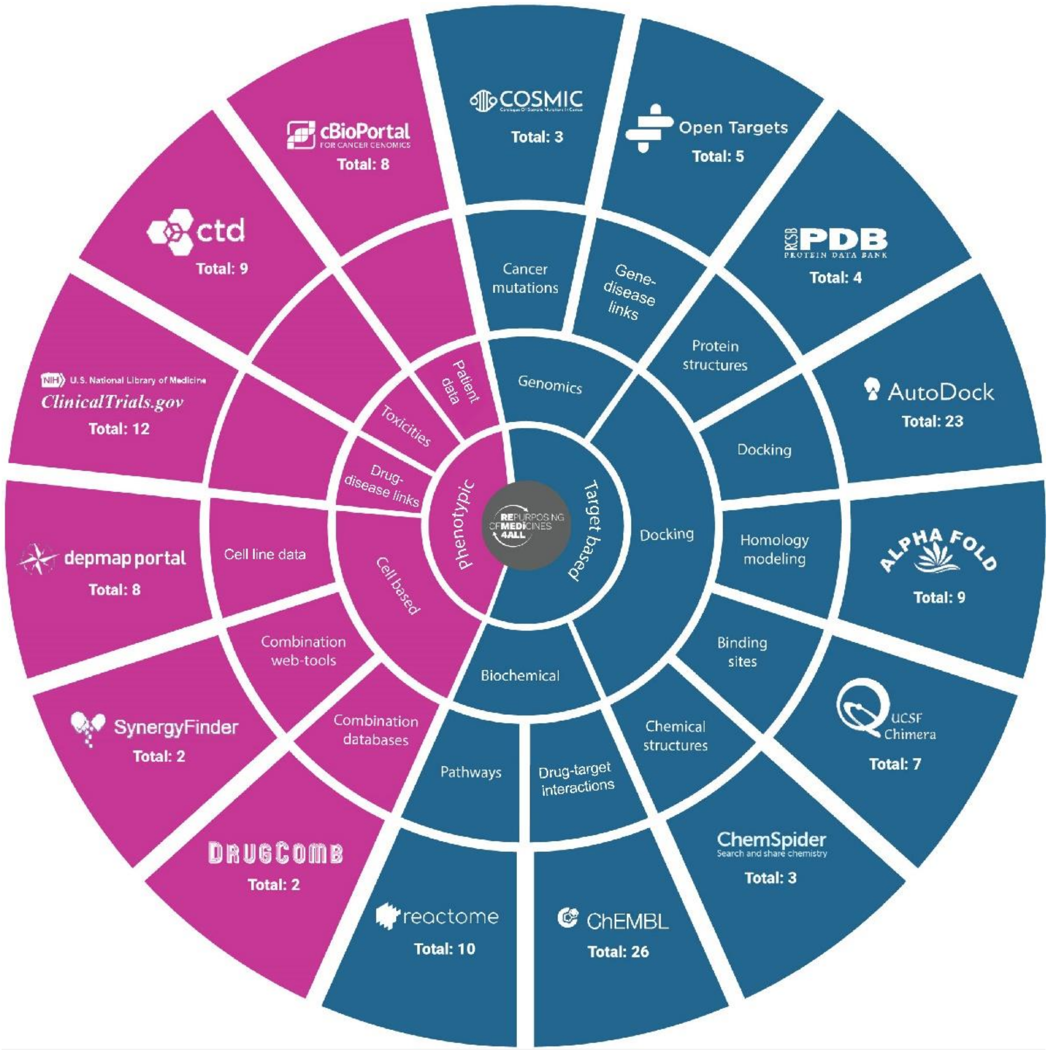

Figure 1: The 196 tools and databases are divided into 15 subcategories of drug repurposing based on the application and data types involved in the tool or database. Each leaf node with dot and lined rectangle shows the DR subtype.

3.2. Top resources in each sub-subcategory

Table 1 shows the top resources' names, types, sub-subcategory names, web-links, median score, and some other features. The first 12 resources are databases, and the remaining four are web-tools. All the 15 resources are open source and had a median ranking score ≥ 8.5 from the experts. Eleven out of the twelve databases come with APIs.

Table 1: The top-ranked resources (databases and tools) in each sub-subcategory.

Resource	Type	Sub-subcategory	Web-link	DR application	Disease / Agnostic	Open source	API	Median score
ChEMBL	Database	Drug-targets interaction	https://www.ebi.ac.uk/chembl/	Drug-target activity prediction on large scale; analysing polymorphic effects	Agnostic	Yes	Yes	10
DrugComb	Database	Combination databases	https://drugcomb.org/	Drug combination prediction	SARS-CoV-2 and cancer	Yes	Yes	9

cBioPortal	Database	Patient data	https://www.cbioportal.org/	Biomarker identification	Cancer	Yes	Yes	9
Dependency Map (DepMap)	Database	Cell line data	https://depmap.org/portal/	Drug response prediction	Cancer	Yes	No	10
ClinicalTrials	Database	Drug-disease links	https://clinicaltrials.gov/	Drug disease relationship prediction	Agnostic	Yes	Yes	10
ChemSpider	Database	Chemical structures	http://www.chemspider.com/	Structure based drug repurposing	Agnostic	Yes	Yes	9.5
Protein Data Bank (PDB)	Database	Protein structures	http://www.rcsb.org	Structure based drug repurposing	Agnostic	Yes	Yes	10
Catalogue Of Somatic Mutations In Cancer (COSMIC)	Database	Cancer mutations	https://cancer.sanger.ac.uk/cosmic	Identify new mutation-cancer associations	Cancer	Yes	No	9
OpenTargets	Database	Gene-disease links	https://genetics.opentargets.org/	Identify new gene-disease associations	Agnostic	Yes	Yes	10
Comparative Toxicogenomics Database (CTD)	Database	Toxicities	http://ctdbase.org/	Drug toxicity assessment	Agnostic	Yes	Yes	10
Reactome	Database	Pathways	http://www.reactome.org/	Predict pathways for the drugs	Agnostic	Yes	Yes	10
SynergyFinder 3.0	Tool	Combination web tools	https://synergyfinder.fimm.fi/	Drug synergy prediction	Cancer	Yes	No	8.5
Autodock vina	Tool	Docking	https://github.com/ccsb-scripps/AutoDock-Vina	Molecular docking	Agnostic	Yes	No	9
AlphaFold	Tool	Homology modelling	https://alphafold.ebi.ac.uk/	Target based drug repurposing	Agnostic	Yes	No	10
Chimera	Tool	Binding sites	https://www.cgl.ucsf.edu/chimera/	Target based drug repurposing	Agnostic	Yes	No	9

3.3. Drug repurposing applications in each sub-subcategory

As mentioned in Section 2.1, the 196 resources were hierarchically classified into 15 sub-subcategories based on drug application and data types. However, some of the resources might use multiple data types and hence can have multiple DR applications. For instance, ‘DrugBank’ is placed under sub-subcategory of Drug-target interactions because it is mainly developed for drug target interactions (DTIs), but it also contains drug-drug interactions (DDI) and pathway information. Hence, other than target hunting, it can also be used for identifying DDIs and pathway mappings. DDI is placed under ‘Phenotypic drug repurposing’. Similarly, some other resources in a sub-subcategory might be developed for oncology applications, while others are disease agnostic.

To get deeper insights, we defined seven further DR classifications, i.e., disease-agnostic, oncology, pathway mappings, design of clinical trials, toxicity assessment, phenotypic DR and target-based DR. We then visualized the proportion of the resources (from each sub-subcategory) that focus on the particular application (Figure 2). As can be seen, most of the resources in different sub-subcategories are disease-agnostic, except for sub-subcategory 'Cancer mutations', where resources are naturally designed for 'Oncology'. Similarly, resources from each sub-subcategory can belong to more than one application out of the seven applications except from 'Toxicities'.

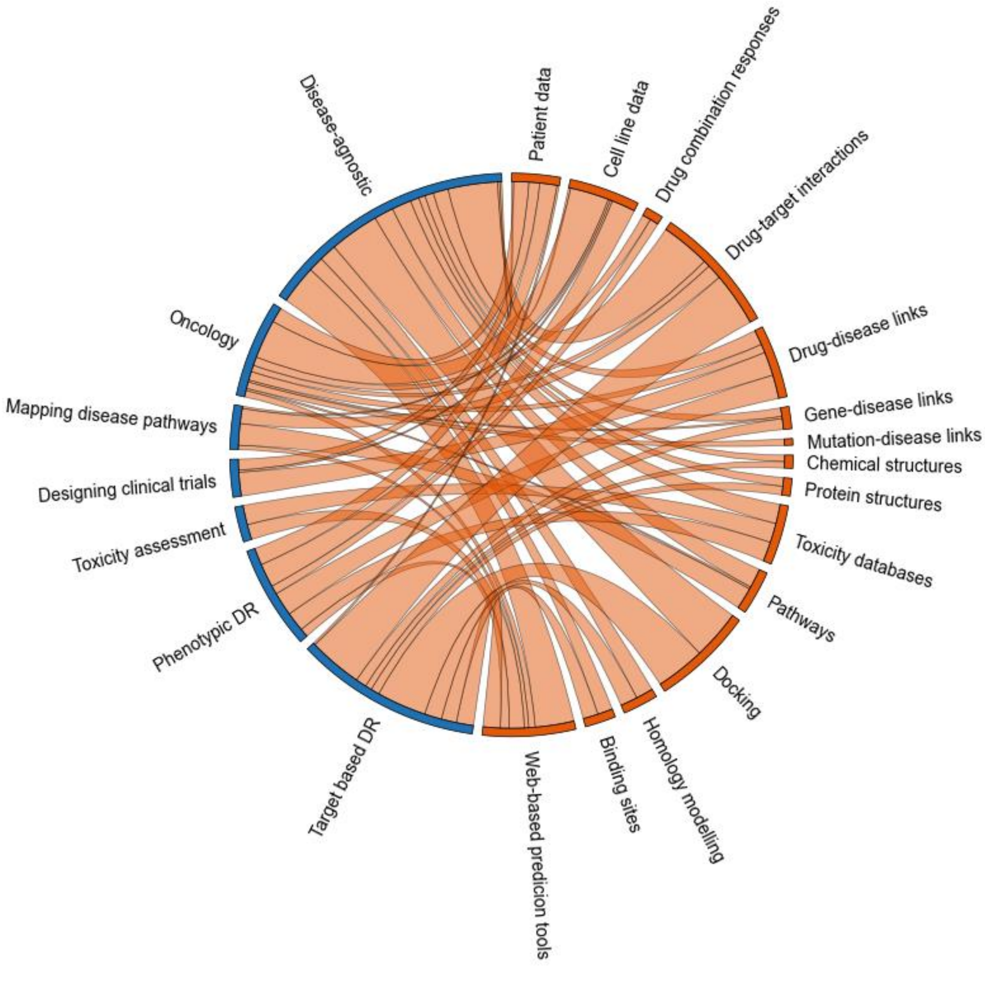

Figure 2: A Chord diagram showing the number of resources for each DR application. Blue color represents DR applications and orange represents different resource categories. Edge widths correspond to the number of resources. This plot can help us to choose the data type or resources for a particular DR application.

3.4. Drug repurposing applications and datatypes surveyed

In addition to databases and tools, Task 4.1 surveyed ten DR prediction methods related to four high-level DR applications. In **Figure 3**, the first layer represents the data types involved in the prediction algorithms, the second layer shows names/details of the methods (journal publications), and the third layer shows the DR applications. The ten prediction methods were mostly based on benchmarking in DREAM challenges or high-impact journal publications comparing several machine-learning methods and the data types for a particular DR application. Coincidently, ten data types were used in the ten methods leading to many-to-many relationships. The selected data types are mutations, copy number variations, gene expression, methylation, chemical properties, drug-target bioactivities, protein/RNA structures, text-based molecular descriptors (based on SMILES, protein sequences or any other text) and clinical features. This plot can help the users to figure out what type of data he/she might need while working on a repurposing application.

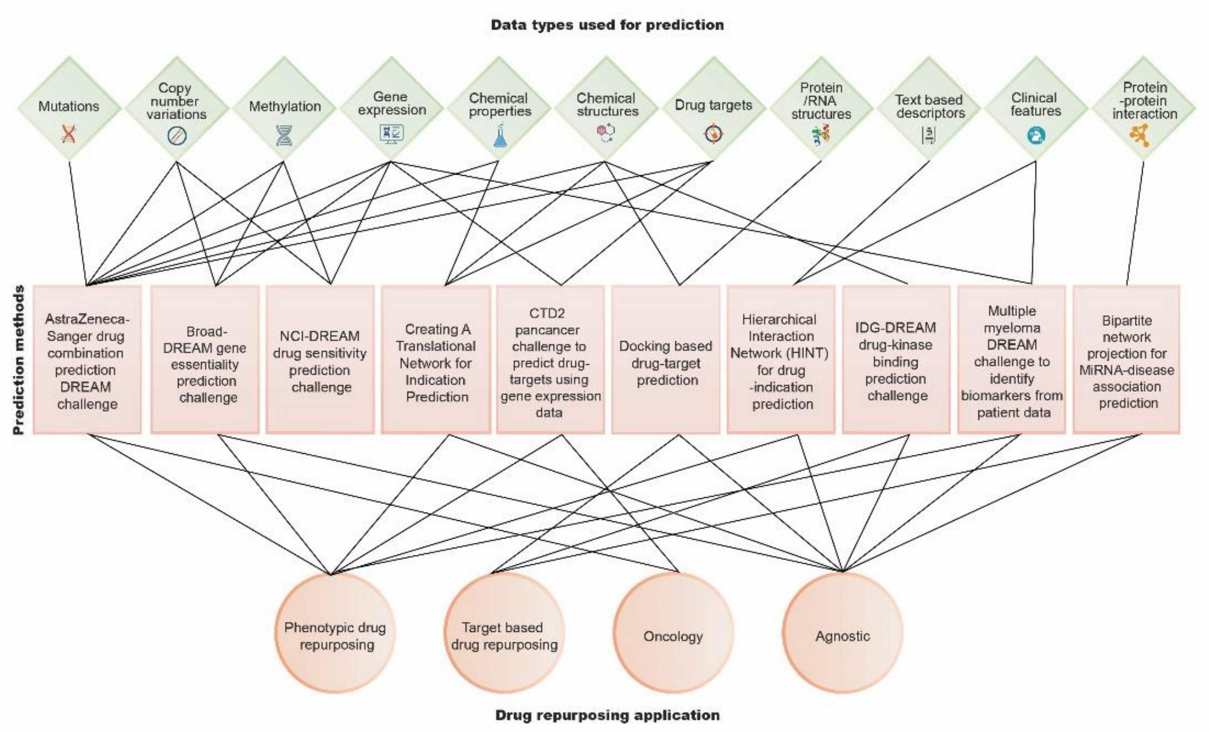

Figure 3: Prediction methods using different data types for four DR applications. Oncology refers to cancer indication, while agnostic means disease-agnostic application.

4. Using *in-silico* resources to support drug repurposing use cases in three demonstrators of the REMEDi4ALL project

4.1. REMEDi4ALL Demonstrators

The tools and processes developed and assembled in REMEDi4ALL are being developed in a portfolio of four ambitious preclinical and clinical phase demonstrators, representing high patient needs in various disease areas, including oncology, rare and infectious diseases. Originally, artificial intelligence (AI) methods were not systematically used to shape the development programme of the demonstrators; instead, the demonstrators were mainly initiated using phenotypic profiling and treatment-related data from the academic lab or pharmaceutical partners. As part of D4.1., we demonstrated the added value of *in-silico* discovery for further repurposing candidate development, and utilized publicly available data from dozens of the surveyed databases and implemented several use cases for three pre-clinical REMEDi4ALL demonstrator projects (Demo 4 is already in clinical phase, so no help needed):

- Demo 1: Drugs combinations related to SARS-CoV-2
- Demo 2: Valproic acid + simvastatin for metastatic pancreatic ductal adenocarcinoma (mPDAC)
- Demo 3: Anti-psoriasis drug (tazarotene) for the ultra-rare disease Multiple Sulfatase Deficiency (MSD)

Full report of the Demonstrator use cases is included at the end of the survey manuscript, which has been sent for publication. Below, we highlight the key analyses and findings from the three use cases. For example, in Demo 2, we show how the knowledge resulted from *in-silico* resources and AI tools can be taken into account to secure patients' safety while defining the study population and introducing safety measures into the clinical trial protocol. The idea behind doing this exercise is to showcase in practice for the researchers how they can use *in-silico* resources to plan and implement future DR studies.

4.1.1. Demonstrator 1: Identify drugs combinations related to SARS-CoV-2

Demonstrator 1 (Demo 1) is based on a phenotypic screening of 5,275 compounds against SARS-CoV-2, using three orthogonal screening assays in Vero E6-infected cells performed at Science for Life Laboratory (SciLifeLab) at Karolinska Institute and Uppsala University, Sweden. A total of 317 compounds were defined as hits based on various selection criteria. Of these hit compounds, 77 were approved drugs, *i.e.*, ready for repurposing, while the rest were either under clinical investigation or classified as preclinical compounds (**Figure 4A**).

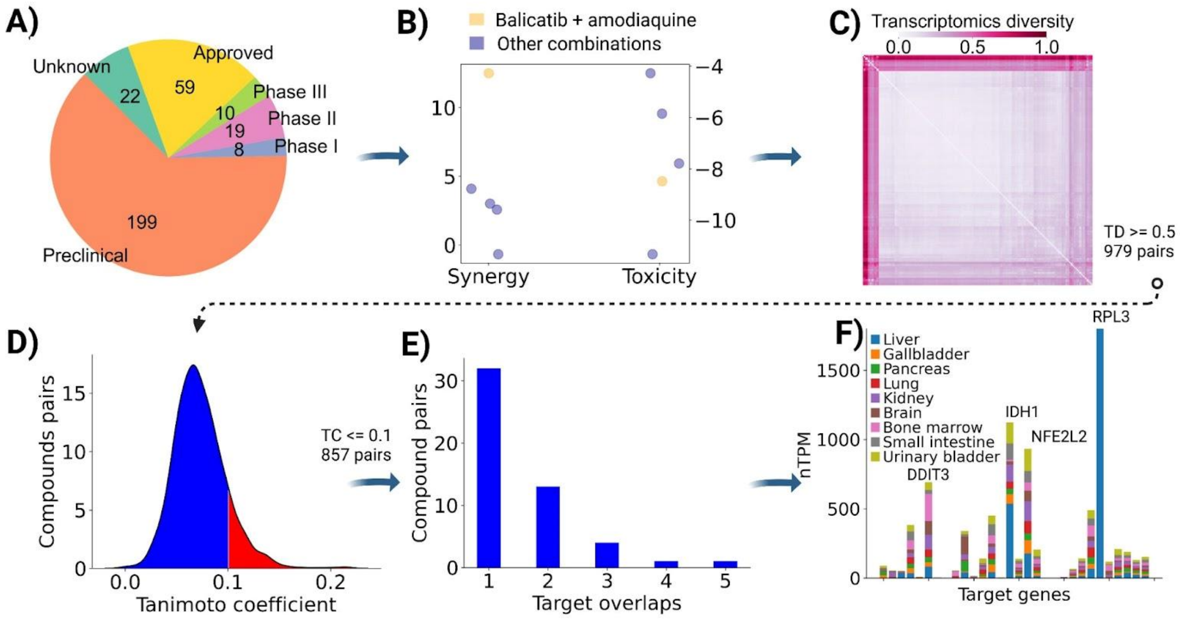

*Figure 4: **A)** Demonstrator 1 is based on 317 hit compounds (both approved and in development), identified in a screening with a DR library (obtained from SPECS), which shows similar fractions of compounds in the development stages. **B)** Identifying synergistic combinations among hit compounds tested in Vero-E6 cells using DrugComb and NCATS databases. Five pairs match with compounds in the hit list, and the balicatib + amodiaquine combination (yellow dots) shows relatively strong synergistic effects (left y-axis) and relatively low toxicity (right y-axis). **C)** Normalized transcriptomics diversity (TD) between the hit compounds using the LINCS L1000 database, and each dot represents TD between a compound pair). A total of 187 hit compounds (17,391 pairs) were matched with compounds in the L1000 database. The gene expression (GE) profiles for the compounds were collapsed together from multiple cell lines at a response time of 24 hours, resulting in the expression profiles for 187 compounds across 12,328 genes. Euclidean distances between each compound pair were calculated and normalized between 0-1. **D)** Structural similarity between 979 compound pairs with high TD (TD>=0.5). We selected 857 compound pairs for which the Tanimoto coefficient (TC) <=0.1 (shown in blue color). **E)** Overlapping drug target profiles of the 857 compound pairs (TC<=0.1), where 51 pairs have shared target profiles (1-5 same target proteins among a total of 32 targets). **F)** Using the Human Protein Atlas database, we analyzed the normalized transcript per million (nTPM) values for the 32 overlapping target proteins across nine SARS-CoV-2 related tissues. Stacked bars represent nPTM values for each tissue, and the names of the top-four most expressed proteins (combined nPTM > 500) are written over the bars.*

The main objective of the Demo 1 use case was to utilize the *in-silico* resources to identify potential combinations among the hit compounds critically required to enhance potency, limit toxicity, and avoid drug resistance (White et al., 2021). To do this, we first explored the DrugComb database to search for combinations already tested in Vero E6 cells exposed to SARS-CoV-2 infection. DrugComb contains 173 such combinations from National Center for Advancing Translational Sciences (NCATS's) SARS-CoV-2 cytopathic effect (CPE) screening data (https://opendata.ncats.nih.gov/covid19/assay?aid=14). Since DrugComb lists only compound names (without compound IDs), we used PubChem's application programming interface (API) to map the compound names into standard International Chemical Identifiers (InChiKeys). After the mapping, we identified five pairs of compounds in DrugComb, where both single agents were present in the Demo 1 hit list (data not shown). Next, we looked into the synergistic and toxicity scores of the five potential combinations from NCATS's matrix combination screening platform (https://ncats.nih.gov/matrix/about). One of the combinations (balicatib+amodiaquine) showed a synergistic potential with a ZIP synergy score higher than 10 (**Figure 4B**, left y-axis). Upon a further inspection of the corresponding host cytotoxicity data (https://opendata.ncats.nih.gov/covid19/assay?aid=15), we found that this combination is also non-toxic to non-infected cells (right y-axis). With relatively high synergy and low toxicity, this combination may potentially boost antiviral potency against SARS-CoV-2.

Since the combination databases provided screening information on only one synergistic combination, we utilized additional *in-silico* resources to identify additional potential combinations among the 317 hit compounds (50,086 possible pairwise combinations).

To understand the biological and chemical space of the 317 hit compounds, we first analyzed their transcriptomics and structural similarities (**Figures 4C and 4D**). One of the largest public omics databases, the LINCS L1000, was used to score the transcriptomics response similarity of the 317 hit compounds. This database contains 3M genome-wide expression response profiles to 34,419 compounds in 273 cell lines (Subramanian et al., 2017). We subset this list to cell lines relevant for SARS-CoV-2 namely, A549, CORL23, H12, NCIH596 and SKLU1 (Asano et al., 2022). Of 317 hit compounds, 19587 contained gene expression profiles in the L1000 database (Level 5) **(Figure 4C)**, leading to a total of 17,391 pairs, out of which 979 pairs have transcriptomics diversity >=0.5. These 979 pairs may have coherence in their mechanism of action (MoA) since they induce similar gene expression changes after compound treatments.

Next, we investigated the structural diversity of these 979 pairs. We computed Extended-Connectivity Fingerprints (ECFP4) using RDKit to analyze the structural similarities (link?). ECFP4 converts the SMILES of each compound into a 1024-bit representation, and the Tanimoto coefficient was used to evaluate the structural similarity between compound pairs. Most of the 979 compound pairs have low structural similarity with the Tanimoto coefficient < 0.25 **(Figure 4D)**. This means that compounds are not just isoforms (or salts) but distinct compounds and can be tested as combinations. We set a relatively stringent threshold of TC<=0.1 (color coded as blue in **Figure 4D**) to pursue efficacious drug combinations, resulting in 857 compound pairs with relatively low transcriptional response similarity and structural diversity.

Compounds with a low structural similarity and high transcriptomic diversity are likely to target diverse protein targets and exhibit versatile MoA. A reasonable hypothesis is that a diverse enough compound library can target different disease-related aspects. In practice, however, to achieve maximal impact on a given phenotype (here, antiviral efficacy), one may need combinations that target pathways that share connected nodes inside the pathogenic “disease module” rather than completely disconnected pathways (Cheng et al., 2019a; Duncan et al., 2012; Tang & Gottlieb, 2022). Such combinations can increase the overall efficacy of the treatment and reduce the risk of drug resistance and lower side effects in cases where synergistic drugs are used at lower concentrations (Jaeger et al., 2017; Yesilkanal et al., 2021). Next, we examined the overlapping target profiles between the 857 compound pairs (**Figure 4D**).

We collected comprehensive target profiles of the hit compounds from six databases: ChEMBL, Drug Target Commons, BindingDB, IUPHAR/BPS Guide to Pharmacology, DrugBank, and DGiDB. To identify potent target activities, we used a target activity cut-off of 1000 nM using multi-dose binding affinity data (Kd, Ki, IC50 and EC50). Compound IDs were mapped to standard InChiKeys using the UniChem database, and the protein targets were mapped to UniProt IDs using the APIs of these databases. There were 51 combinations among 51 unique compounds for which we could find shared target proteins (**Figure 4E**).

We hypothesized that the compound pairs targeting the top-expressed gene targets could identify new potential combination therapies for SARS-CoV-2. However, compounds with overlapping binding sites for a shared target protein, when combined, could lead to unwanted side effects or toxicities if the target proteins are not related to the disease being treated. We next investigated the gene expression profiles of the 32 targets using the Human Protein Atlas (HPA) database across six SARS-CoV-2-related tissues (**Figure 4F**). Four out of 32 target genes (RPL3, NFE2L2, IDH1 and DDIT3) showed high expression levels and seven drug combinations out of 51 pairs shared any of these four targets. The top-expressed gene, 60S ribosomal protein L3 (RPL3), is a shared target for cycloheximide and anisomycin; DNA damage-inducible transcript three protein (DDIT3) is shared by curcumin and bortezomib, and isocitrate dehydrogenase (IDH1) is shared by disulfiram and anisomycin. Four combinations have Nuclear factor erythroid 2-related factor 2 (NFE2L2) as the shared target, i.e., sulforaphane and bortezomib, disulfiram and bortezomib, simvastatin and bortezomib, and cycloheximide and bortezomib.

Finally, we wanted to see whether new language models could identify the same or additional combinations for SARS-CoV-2. Using ChatGPT-4, we searched the literature for any studies focusing on the drug combinations sharing these four targets. Most of the literature references returned by ChatGPT were wrong; however, it identified one correct reference (Chen et al., 2010). It is expected that the next versions of ChatGPT will be improved for biological use cases. The literature search indicated that our predicted combinations have not been previously reported in the SARS-Cov-2 context. However, one of these seven suggested combinations (sorafenib and bortezomib) is already tested in preclinical experimentation for the treatment of hepatocellular carcinoma (Chen et al., 2010) and advanced malignant melanoma (Sullivan et al., 2015). Our exercise identified seven drug combinations to be validated *in-vitro* and *in vivo* to indicate SARS-CoV-2. Two of these, bortezomib + disulfiram and simvastatin + bortezomib, are based on approved drugs and could be promising repurposing candidates for treating SARS-CoV-2.

4.1.1. Demonstrator 2: Valproic acid-simvastatin combination for pancreatic cancer

Demonstrator 2 (Demo2) is based on the assessment of the synergistic antitumor interaction between simvastatin and valproic acid and its potentiation of the standard chemotherapy (Gemcitabine + Nab-paclitaxel) in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC). This synergistic action has been demonstrated in preclinical studies (Iannelli et al., 2020a). Compared to chemotherapy alone, a clinical trial to test the new combinatorial therapy's efficacy has recently started recruiting patients (https://clinicaltrials.gov/ct2/show/NCT05821556). Multi-omics studies in vitro and in vivo are underway to better understand the mechanism of action of this drug combination.

To explore the known primary and secondary targets of valproic acid and simvastatin, we used different databases that contain data about chemo-biological interactions from scientific articles and patents, namely ChEMBL, DrugBank, BindingDB, ChemicalChecker and Therapeutic Target Database (TTD). In addition to the known targets retrieved from these databases, we also explored predicted targets based on similarity and machine learning (ML) approaches for repurposing with the CLARITY technology platform (https://chemotargets.com/clarity/), developed by the REMEDi4ALL partner Chemotargets. From these resources, we gathered 34 potential targets for simvastatin and 64 for valproic acid (**Fig. 5A**). We found 15 targets to be in common between the two drugs. Notably, most of the shared targets comprise enzymes annotated in DrugBank, namely cytochromes (e.g., CYPs), esterases (e.g., CESs) and transferases (e.g., UGTs), among others (**Fig. 5B**).

Mapping of the shared targets to the KEGG pathway database revealed 35 biological processes grouped in 4 main categories, from which the metabolism of xenobiotics and endobiotics was the most populated (**Fig. 5C**). The prevalence of common targets belonging to this pathway implies that care must be taken to study the pharmacokinetics of the two drugs, when administered together. Indeed, one of the common targets (UGT1A1) was also reported by DrugBank's Drug Interaction checker tool as a common substrate that may be responsible for blocking the metabolism of simvastatin when combined with valproic acid (**Fig. 5C**) (Marques & Ikediobi, 2010; Ping Goon et al., 2016; Zhang et al., 2015).

The relatively low number of common pharmacodynamic targets suggests that these two drugs may exert a synergistic action by simultaneously hitting different proteins. Often, drug synergy arises from the complementary perturbation of the same disease pathways, an observation that has been exploited to design drug combinations rationally (Cheng et al., 2019b; Zhou et al., 2020). To further explore this possibility, we also annotated the KEGG pathways for the protein targets that were not in common but unique to the two drugs. We found 26 shared pathways that involved at least one target for simvastatin and one for valproic acid (**Fig. 5D**). Interestingly, among these, the AMPK signalling pathway has been previously reported to be modulated by the combination of valproic acid and simvastatin (Iannelli et al., 2020b), thus supporting our in silico results.

When studying the MoA, one must consider that the actual active compound may not necessarily be the drug but one of its metabolites. Therefore, it is advisable to also inspect the drug's available metabolites, for example, from the ChEMBL's compound report cards (e.g. CHEMBL109 (https://www.ebi.ac.uk/chembl/compound_report_card/CHEMBL109/) and CHEMBL1064 (https://www.ebi.ac.uk/chembl/compound_report_card/CHEMBL1064/) for valproic acid and simvastatin, respectively). These metabolites can be used as drug activity surrogates when measured from patient samples treated with the drug combination in clinical trials, which may shed light on the drug's mechanism of action and toxicity. Additionally, the targets of these metabolites could be further explored to identify new potential drug interactions. Altogether, the exploration of the diverse *in-silico* resources assembled by REMEDi4ALL paves the way for the thorough identification of drug associations, covering distinct yet relevant phases of the drugs' biological action, such as their pharmacodynamics and pharmacokinetics.

To explore the safety events reported for patients taking valproic acid, simvastatin, or both, we explored the post marketing reports of adverse events from four databases (FAERS, JADER, VAERS and Vigibase), which have been standardized and de-duplicated in the CLARITY PV technology platform from REMDi4ALL partner Chemotargets (https://claritypv.com/). To identify statistically significant events, we relied on the proportional reporting ratio (PRR (Sakaeda et al., 2013)), a statistic that measures the extent to which an adverse event is more frequently reported with a specific drug than with the other drugs. More specifically, the reports were prioritized by safety events that showed a PRR higher than 2, either for valproic acid or simvastatin, qualifying these safety signals as Adverse Drug Reactions (ADRs). Reassuringly, the PRR values are not increased in patients taking both valproic acid and simvastatin when compared to monotherapy use of the drugs (**Fig. 5F**). This is important, particularly to address concerns raised by the common targets analysis of the combination treatment. In fact, despite the possible metabolic interactions, no signs of increased toxicity are evident in the population of patients taking both these drugs.

As an example, valproic acid shows an ADR signal for pancreatitis (PRR higher than 2) in post-marketing reports, which was already reported in the approved label for valproic acid (https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=002d4c7e-b94e-49b6-8554-0aa0c60d5bde). Conversely, while some reports link pancreatitis to simvastatin, it is not a statistically significant ADR signal according to CLARITY PV. In fact, the PRR value for pancreatitis is slightly lower in the patients taking both valproic acid and simvastatin than in those patients taking valproic acid alone. Overall, the exploration of postmarketing reports offers a systematic means for early detection of those ADRs that may be exacerbated by the drug combination, hence anticipating the risk of failure due to increased toxicity in subsequent clinical stages.

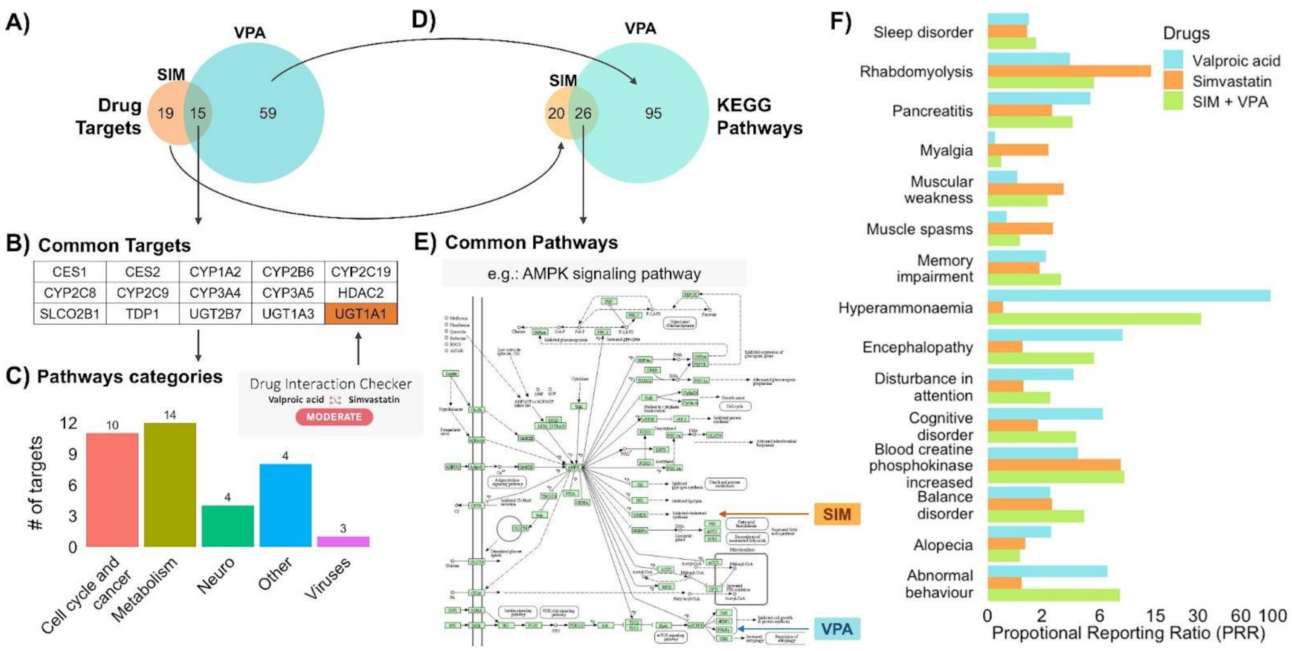

Figure 5: ***A)*** *Number of unique and common targets retrieved from five curated databases and one predictive tool (Clarity).* ***B)*** *15 common targets for simvastatin and valproic acid and their interaction results from DrugBank's Drug Interaction Checker.* ***C)*** *The number of common targets in each main KEGG pathway category. The number of pathways per category is reported on top of each bar.* ***D)*** *Venn diagram of KEGG pathways annotated to the targets, not in common.* ***E)*** *26 common pathways for the unique targets of simvastatin and valproic acid. An example pathway of AMPK signalling.* ***F)*** *Proportional Reporting Ratios (PRR) for the main safety events that show Adverse Drug Reaction signals from patients taking either valproic acid, simvastatin, or their combination (all data shown in the supplementary material). Note: the x-axis is plotted in logarithmic scale.*

4.1.3. Demonstrator 3: Anti-psoriasis drug (tazarotene) for the ultra-rare disease Multiple Sulfatase Deficiency (MSD)

Demonstrator 3 (Demo 3) is based on the discovery of Tazarotene, an approved drug for the topical treatment of psoriasis and acne (Weinstein et al., 1997), as a repurposing candidate drug for the treatment of multiple sulfatase deficiency (MSD) (Schlotawa et al., 2023). MSD is characterized by a global deficiency in cellular sulfatases. This deficiency occurs by a lack of proper sulfatase post-translational activation, which is usually carried out by the formylglycine-generating enzyme (FGE) (JJ, 2001). Therefore, a biochemical screen was performed using an optimized arylsulfatase A (ARSA) assay to identify drugs to rescue defective sulfatase function using patient-derived fibroblasts and a library of 785 approved drugs. Tazarotene and Bexarotene were found to increase ARSA activity and were subsequently shown to rescue specific MSD cellular phenotypes in both fibroblasts and neuronal progenitor cells (NPC) (Schlotawa et al., 2023). Tazarotene and Bexarotene are known retinoids that respectively target retinoic acid receptors RAR and RXR. Despite these canonical targets and the fact that Tazarotene and Bexarotene seemingly stabilized misfolded FGE proteins, which in turn increased sulfatase activities rescuing the MSD phenotype, the molecular mechanism and potential other targets of Tazarotene and Bexarotene remain unclear.

Thus, the main objective of the following demonstrator was to investigate how available *in silico* tools can aid in elaborating data-driven hypotheses for this particular case. For clarity and simplicity purposes, we focused solely on Tazarotene, given its demonstrated superior rescuing effect than Bexarotene in MSD disease models (Schlotawa et al., 2023). The main objective of this use case was to utilize available data to a) elucidate a potential mechanism by which Tazarotene rescues the MSD phenotype and b) identify potential drug combinations that could lead to a positive treatment outcome. To identify other targets than those considered canonical, which could explain potential mechanisms of action, we used three databases, namely ChEMBL, Drug Gene Interaction database (DGIdb) and Drug Target Commons (DTC) (**Figure 6A**). Interestingly, even without any potency cut-offs, only three common targets were identified using the bioactivity data in the three databases, corresponding to the canonical retinoic acid receptors RARA, RARB and RARG. When combining the shared targets found in at least two databases, additional five targets appeared, i.e. USP1, KMT2A, EHMT2, GMNN and NFE2L2 (**Figure 6B** upper panel). There was a larger number of unique targets in each database, presenting a potential list of alternative targets (**Figure 6B** lower panel).

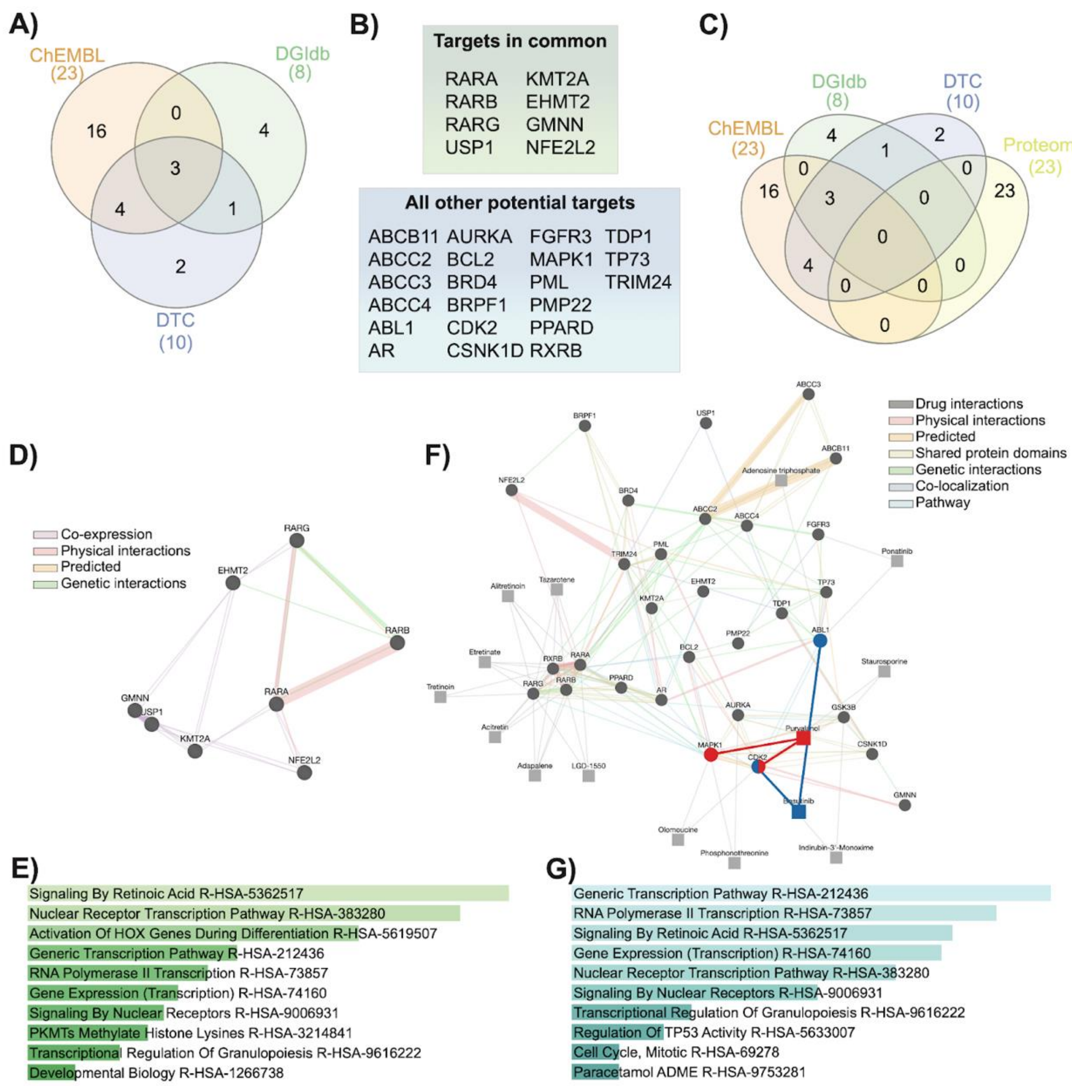

***Figure 6:** Target, network and pathway analysis for Tazarotene's potential mechanism of action elucidation. **A)** Venn diagram for common and unique targets for Tazarotene extracted from ChEMBL, Drug Gene Interactions database (DGIdb) and Drug Target Commons (DTC). **B)** List of the common targets among the three databases used in A (upper panel). List of the potential targets found common in two or three databases used in panel A (lower panel). **C)** Venn diagram including all Tazarotene's targets extracted from ChEMBL, DGIdb, DTC and significantly up/down regulated proteins upon Tazarotene treatment in HT116 cells (Mitchell et al., 2023a). **D)** Genemania network analysis, including the targets reported in common between ChEMBL, DGIdb and DTC. **E)** Reactome pathway enrichment analysis using the common targets extracted from ChEMBL, DGIdb and DTC using Enrichr. **F)** Genemania network analysis, including all the reported targets from ChEMBL, DGIdb and DTC (common and unique targets). **G)** Reactome pathway enrichment analysis using all the reported targets extracted from ChEMBL, DGIdb and DTC.*

Proteomics and transcriptomics are among the most commonly used omics technologies to identify differentially expressed molecules, in this case proteins, upon a given stimulus or treatment. These genome-wide approaches can also be leveraged towards identification of a drug mechanism of action. In a recent publication, 875 compounds were profiled using quantitative proteomics in the colon cancer cell line HCT116 (Mitchell et al., 2023b). Tazarotene was one of the drugs profiled in this study at a concentration of 10µM, the same concentration found to efficiently rescue the MSD phenotype in fibroblasts (Schlotawa et al., 2023). The mass spectrometry-based global proteomics approach identified 23 up and down-regulated proteins upon Tazarotene treatment in HCT116 cell line. Unexpectedly, none of the up/down-regulated proteins coincided with the described canonical or potential Tazarotene targets found in the three databases (**Figure 6C**). This suggests either a context-specificity of the Tazarotene inhibition mechanisms, specific to HCT116 cells, or that the treatment effect is modulated through other off-targets of the drug, not yet reported in any of the databases.

We took a network approach to explore further the potential mechanism behind Tazarotene's rescue effect on MSD. We used the Genemania tool (Warde-Farley et al., 2010) to visualize interactions among the canonical targets of Tazarotene (**Figure 6D**). We allowed Genemania to use co-expression, physical interaction, genetic interaction, and prediction data to build the network. The majority of the interactions were based on physical (51.6%) and co-expression (23%), followed by genetic interactions (20.4%) and predicted interactions (5%), which we considered to have the least contribution to the MoA of the treatment effects. Next, we investigated the pathways enriched by the canonical targets of Tazarotene using Enrichr (Kuleshov et al., 2016), a comprehensive database for pathway enrichment analysis. Among the multiple pathway databases that Enrichr offers, we used Reactome 2022 reference (**Figure 6E**). The pathway enrichment analysis returned retinoic acid signaling as the main pathway, but also transcription related pathways, such as Nuclear Receptor Transcription and RNA Pol II transcription, indicating a potential transcription regulation by Tazarotene.

To explore the possibility of identifying alternative pathways that would explain Tazarotene's mechanism of action, we again took a network approach. This time, inputting all the potential targets identified in the three databases (ChEMBL, DGIdb and DTC) in Genemania resulted in a larger interconnected network **(Figure 6F**). Interestingly, the output included drugs that targeted several of the potential alternative targets (**Figure 6F**, blue and red highlighted drugs (squares) and their corresponding targets (circles)). Finally, we used Enricher to investigate the additional pathways that appeared to be involved when using all the potential targets as a query, revealing again pathways related to transcription (**Figure 6G**) and indicating a potential transcriptional modulation by Tazarotene.

Several conclusions can be drawn from this *in silico* experiment. The fact that only Tazarotene's canonical targets were identified as the common ones out of three databases indicates that a careful interpretation of the target activity data is needed, including an in-depth understanding of how these data were obtained, to discard weak interactions. To our surprise, none of the significantly up/down regulated proteins identified by global proteomics corresponded to the canonical or non-canonical Tazarotene's targets. The different disease models could explain this lack of concordance (colon cancer HT116 vs MSD primary fibroblasts). However, it once again highlights the importance of cautiously interpreting the data. Our network and pathway analysis pointed to transcription being regulated, at least when using Tazarotene's canonical and non-canonical targets as inputs. In addition, a few drugs, such as Purvalanol, a cyclin-dependent kinase 2 (CDK2) (Villerbu et al., 2002) and mitogen activated kinase (MAPK1) inhibitors (Ivanchuk & Rutka, 2006), and Bosutinib, a BCR-ABL (Golas et al., 2003) and CDK2 inhibitor (Mancini et al., 2007), seemingly targeting some of the connected potential targets of Tazarotene, could offer the possibility for a combinatorial treatment by which several pathways would be targeted. In conclusion, our results demonstrate the usefulness of existing *in silico* tools to generate data-driven hypotheses and potential to facilitate new discoveries, which will need further experimental validation.

5. Conclusion

Together with REMEDi4ALL partners and external experts, we surveyed 196 tools, databases, and methods for DR applications. We divided the resources into subcategories based on data types and DR applications for which resources could be applied. Detailed annotations for these 196 resources have been made available for the community online at https://idrc-r4a.com/. Such a sustainable and extendable web-repository of *in-silico* DR resources will help researchers to choose the right resource for a given DR application. We also showed different data types in the ten surveyed prediction methods for a particular DR application (**Figure 3**). This can help to understand what types of data are needed for a particular DR prediction method. Finally, we utilized three (of four) demonstrator projects for REMEDi4ALL to showcase the usage of *in-silico* resources in developing DR pipelines.

Cataloguing of *in-silico* resources for various drug repurposing use cases is expected to help in various DR applications, such as disease/target/candidate/therapy identification and lead hypothesis generation. While earlier evaluations have been based on other types of criteria, we emphasized in Task 4.1 the practicality and usefulness of the resources in the expert-based evaluation of top-resources in each use case category. The evaluation criteria included both objective factors, such as data statistics, FAIR principles and frequency of the updates, as well as subjective criteria, such as data/resource confidence, type of drug-repurposing applications and whether the resource is disease-specific or -agnostic. The experts also indicated potential improvements in the existing resources, and limitations of the existing resources (e.g., many resources are focused on oncology applications).

While a more formal gap analysis will be implemented as part of D4.2., when implementing the survey and demonstrator use cases, we noticed that the chemical ID mapping system needs to be further improved, using e.g. UniChem ID mappings that are currently applied only in 30 of the surveyed chemical databases. The lack of unique ID mappings limits integrative analyses that require combining data from multiple data resources, e.g., compound sensitivity profiles with drug-target bioactivity datasets. In comparison, UniProt is being used in almost all the protein target-related databases. Similarly, the clinical data resources need to be better structured and more frequently updated. For instance, SIDER, one of the most-cited databases for adverse drug effects was last time updated in 2015 (as per the Sider website).

Furthermore, https://clinicaltrials.gov/ should be annotated with drug and disease identifiers to improve its usability, and allowing for linking clinical trial data with other data types such as gene-disease associations. Since the ClinicalTrials database provides API for a programmatic access, computational researchers could develop automated systems that link drugs and diseases in ClinicalTrials with respective identifiers such as PubChem ID for compounds and Mesh IDs for the diseases. This is expected to reduce redundancy while processing the ClinicalTrials datasets.

The expert evaluations and use cases also indicated other potential improvements and limitations of the existing resources. Notably, among the 100 surveyed databases, only 39 currently implement APIs. APIs are important for cross-linking and for FAIR sharing of the data, making it critical for the authors to provide APIs in the future releases of their databases. This will improve the re-usability of the data and open new opportunities for data-driven DR applications. Furthermore, several databases provide useful statistics on their website (e.g., number of compounds, cell lines, targets, diseases, or interactions); however, extracting such statistics for the remainder of the databases was a tedious task, and we had to read several articles to get up-to-date statistics for each database. We suggest the authors of the databases specify these details on their websites for future applications, so that these can be updated in the next version of our DR resource catalogue (https://idrc-r4a.com/).

We hope this work provides a basis for future extension of the *in-silico* DR catalogue, and we will see more targeted *in-silico* solutions for specific DR applications, in addition to the broadly applicable resources that are currently used in most of the DR studies (e.g. the top-15 resources in the current evaluation). While there are several *in-silico* methods supporting 'off-target' drug repurposing by linking drugs to disease mechanisms, there remain need for methods to identify new links between indications and mechanisms, to support 'on-target' repurposing, wherein the drug works via the same mechanism as in its original indication. Novel *in-silico* platforms are needed also for predicting drug combination effects, both therapeutic and adverse reactions, and for making available and cross-linking clinical patient-level data for treatment outcomes.